



NeuroQueue

The scan that cannot wait gets read first.

AI-assisted brain MRI triage and reporting. Our trained model sorts the queue, a doctor reviews and signs every scan, and the patient receives a clear report.

TEAM

Import Innovation

LIVE PRODUCT

neuroqueue-one.vercel.app

MEMBERS

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Sanmeet · Mohamed Abubakker

SOURCE CODE

github.com/ABHIRAM5744/NeuroQueue

Decision support only, not a diagnosis. Every scan is read by a doctor.

The project in one sentence

We help radiology departments, their doctors and the patients waiting on them solve urgent brain MRI scans waiting in a first-come queue while every report is written by hand through NeuroQueue, a real-time triage and reporting web platform and our own trained MRI classifier with a rule-based safety verifier, plus an LLM that only words the report, improving time-to-read for urgent scans to a simulated mean wait of 6.5 minutes instead of 17.8, with no increase in total waiting.

98.67%

Test accuracy
on 901 scans the model
never saw

11 of 12

Model errors still
caught
routed to Urgent or
Review by the verifier

6.5 min

Wait for a tumour scan
instead of 17.8 (simulation)

Live

Deployed product
3 portals · English and
Arabic · 87 tests

What this document covers

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Purpose, team, core value

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Where to see it

LIVE PRODUCT

neuroqueue-one.vercel.app

RECORDING (78 S)

[NeuroQueue-launch-video.mp4](#)

SOURCE CODE

github.com/ABHIRAM5744/NeuroQueue

01

Project Summary

Purpose, team, core value

PURPOSE

Make sure the brain MRI scan that cannot wait is read first, and give the doctor who reads it a finished draft report instead of a blank page.

WHAT IT IS

A deployed web application with three portals (administrator, doctor, patient), live updates on every screen, in English and Arabic.

Core value: three jobs that never blur

Our trained model classifies

It is the only thing that classifies a scan: glioma, meningioma, pituitary tumour or no tumour.

The doctor decides

The assigned doctor reviews every scan and signs every report. No scan is ever marked done by software.

The language model only writes

Qwen words the report from the doctor's confirmed fields. It never sees the image and never classifies.

Three portals

PORTAL	WHAT THE PERSON DOES
Administrator	Uploads one or many MRI scans, assigns each batch to a doctor and a patient, approves doctor registrations, oversees the system
Doctor	Sees only the scans assigned to them, reviews the classification with the evidence, confirms or overrides, approves and signs the report
Patient	Sees and downloads only their own doctor-finalized report, in plain language

02

Problem & Users

Who struggles, and why

A tumour scan waits behind every normal scan that arrived before it. Brain MRI scans are usually read in arrival order, the report is typed by hand, and the patient receives a document written for clinicians. How long an urgent scan waits depends on luck.

DOCTORS

One long, undifferentiated worklist with no signal about which scan to open first, and every report typed from scratch.

CLINIC ADMINISTRATORS

No single place to route scans to the right doctor, see what is unread, or prove who did what.

PATIENTS

Long waits with no visibility, then a report full of clinical language.

03

Solution & Core Flow

The core scenario, from input to visible result

1

Admin uploads and assigns

One image or up to 50 at once. The admin chooses the doctor and the patient, which links them.

2

The trained model classifies

Four classes with a calibrated confidence. Starts automatically after upload.

3

A rule-based verifier sets the tier

Urgent **Review** **Routine** with the reason shown. Over 60 minutes waiting escalates to Urgent.

4

The assigned doctor reviews

Image, probability of every class and a heatmap of where the model looked. Confirm, or override with a reason.

5

The report drafts itself

A clinical report and a plain-language finding, written only from what the doctor confirmed.

6

Six checks must pass

A tumour type, size or location that differs from the review, or any malignancy wording, blocks signing.

7

The doctor signs

Two PDFs are created: a clinical report and a separate plain-language patient report.

8

The patient receives it

The patient portal shows only their own finalized report.

Every step updates live over a WebSocket and is written to a tamper-evident audit log. Also included: Arabic interface with right-to-left layout, a voice help assistant, email verification, password reset and Google sign-in.

04

AI, Data & Tools

Models, sources, stack, and exactly where each is used

Which model runs where, and how it is called

MODEL	WHERE IT IS USED	HOW IT IS CALLED
EfficientNet-B0, trained by us	Classifies every uploaded scan. The only classifier	Runs inside our backend as ONNX. No outside service sees the image
Deterministic verifier (rules, no AI)	Turns probabilities and waiting time into a tier with a stated reason	Plain code. It can never mark a scan as cleared
Qwen3.8-Max	Drafts the report wording after the doctor's review; translated the interface into Arabic	Alibaba Cloud Model Studio, from the backend only. Receives the doctor's confirmed fields, never the image
Qwen3.8-Flash	Help assistant answers; plain-language "why this tier" explanation	Alibaba Cloud Model Studio, from the backend only, streamed
AssemblyAI streaming	Voice input for the help assistant	Browser streams audio with a 5-minute token from our backend; the key stays on the server

Measured on 901 held-out test images

MEASURE	RESULT
Accuracy	98.67%
Tumour recall	99.1%
Misclassified scans	12 of 901
Of those, still sent to Urgent or Review	11 of 12
Calibration error (ECE)	0.004 (0.008 before scaling)

DATA

Brain Tumor MRI Dataset (Kaggle, M. Nickparvar, CC0). 7,200 images indexed. **1,144 duplicates and 49 label conflicts removed before splitting**, so no image is in both training and test. 6,007 kept, split 70 / 15 / 15. Thresholds tuned on validation only.

Recall by class



Scale starts at 90%.

STACK

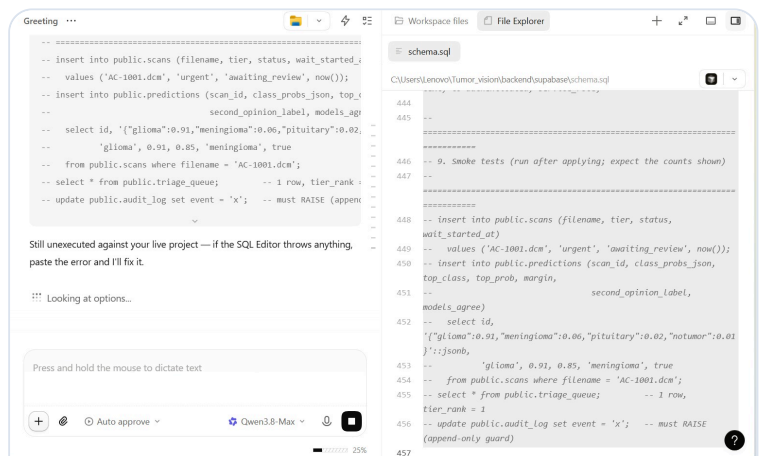
Next.js 15, React 19, Tailwind, three.js on Vercel · FastAPI, ONNX Runtime, ReportLab on Railway · Supabase Postgres and Storage · WebSockets · PyTorch for training · 87 automated backend tests.

Built with Qoder

We used **Qoder**, with the **Qwen3.8-Max** model selected, for two parts of the build:

- **Animation integration:** building the interface animations into the web application
- **SQL queries:** some of the queries for the database schema

The screenshot is from one of those early development sessions.



PROOF A Qoder session with Qwen3.8-Max selected, working on our database schema SQL.

05 Demo Evidence

Grouped screenshot sequence of the complete core flow, taken from the live site

RECORDING

78-second video of the full flow:
[NeuroQueue-launch-video.mp4](#)

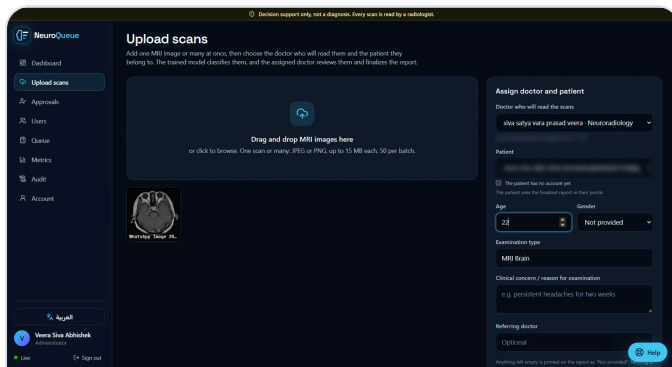
TRY IT

[neuroqueue-one.vercel.app](#)

SLIDES

[NeuroQueue-Pitch-Deck.pptx](#)

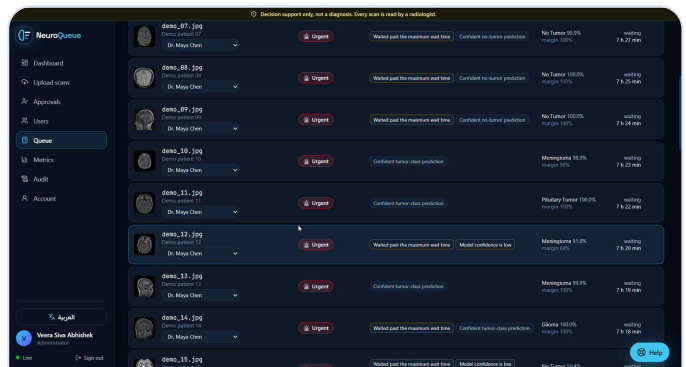
A · INPUT: THE ADMINISTRATOR UPLOADS AND ASSIGNS



STEP 1

Upload and assign

One image or many. The admin chooses the doctor and the patient.

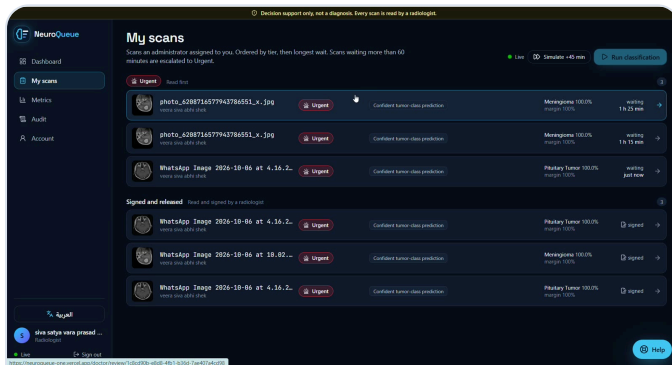


STEPS 2-3

Classified and tiered

The trained model has classified each scan; the queue is ordered Urgent, Review, Routine.

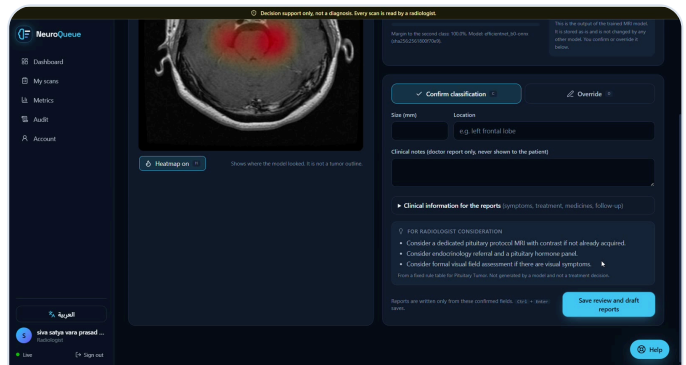
B · THE ASSIGNED DOCTOR REVIEWS AND THE REPORT IS PRODUCED



STEP 4

Only assigned scans

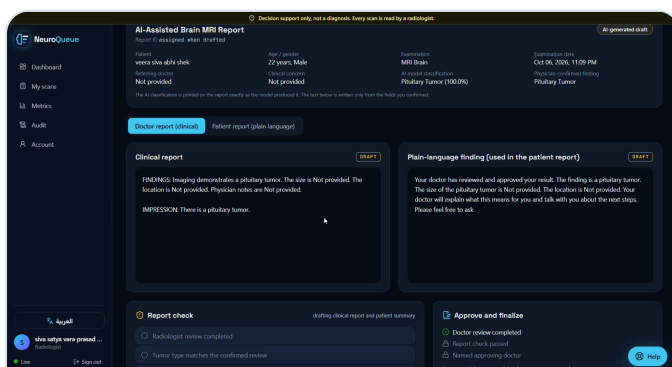
The doctor signs in and sees only the scans assigned to them.



STEP 4

Review with evidence

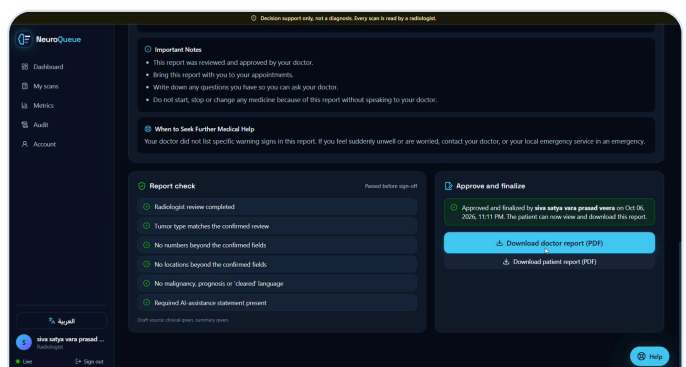
Image, class probabilities and heatmap; confirm or override.



STEP 5

The report drafts itself

Generated from the doctor's confirmed fields when the review is saved.

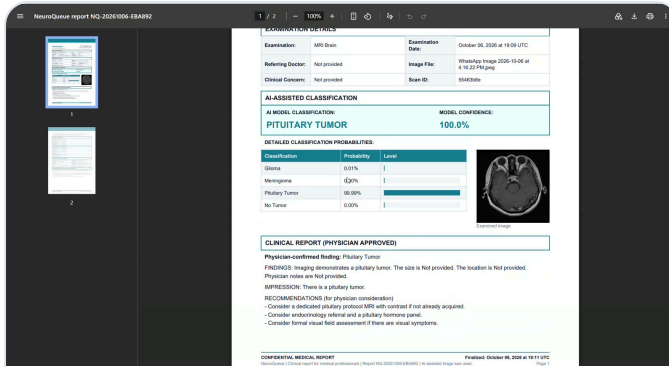


STEPS 6-7

Checks passed, report signed

All six check rows passed; the doctor has approved and finalized.

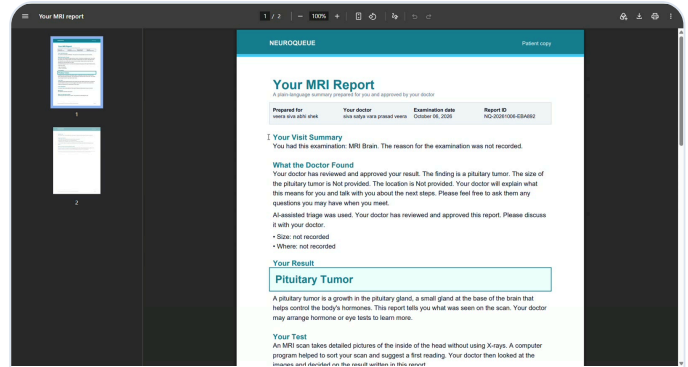
C · VISIBLE RESULT: TWO REPORTS AND THE PATIENT PORTAL



STEP 7

Clinical PDF for the doctor

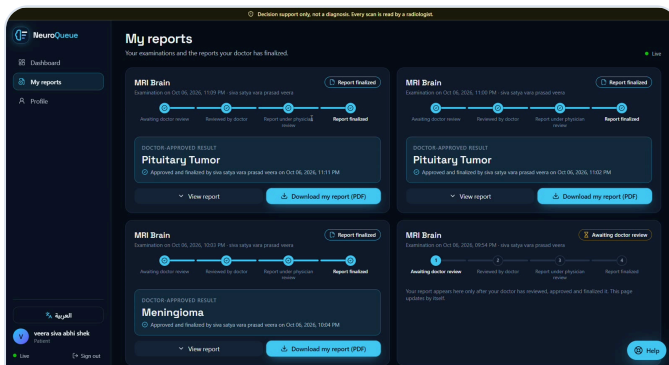
Model classification, probabilities and physician-approved findings.



STEP 7

Plain-language PDF for the patient

No probabilities, tiers or model internals.



STEP 8

Patient portal

The patient sees only their own finalized reports.

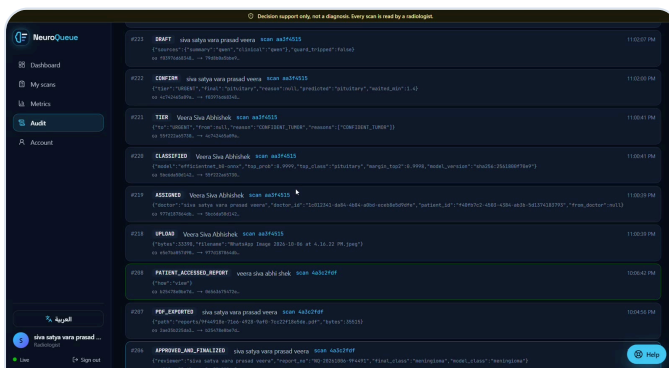


PROOF

Model metrics page

Accuracy, recall, errors caught by the verifier, calibration, confusion matrix.

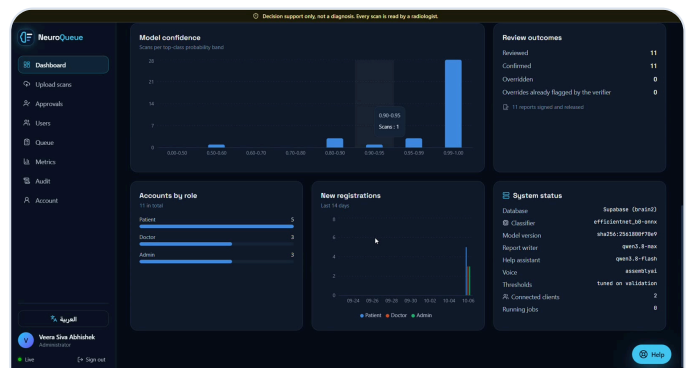
D · PLATFORM FEATURES



AUDIT

Tamper-evident audit log

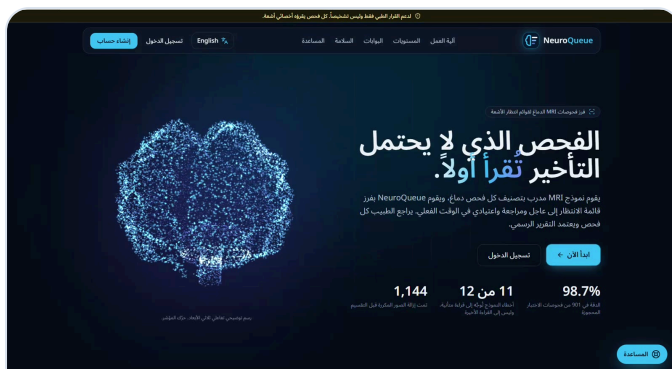
Every upload, assignment, classification, review and approval is hash-chained.



LIVE

Administrator dashboard

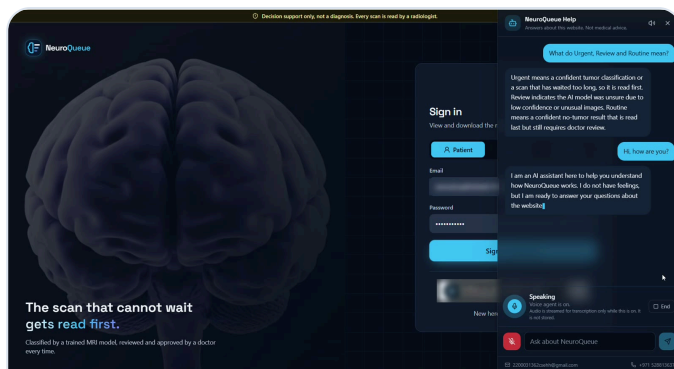
Every number updates live.



LANGUAGE

English and Arabic

The whole site switches language and direction with one button.



VOICE

Help assistant

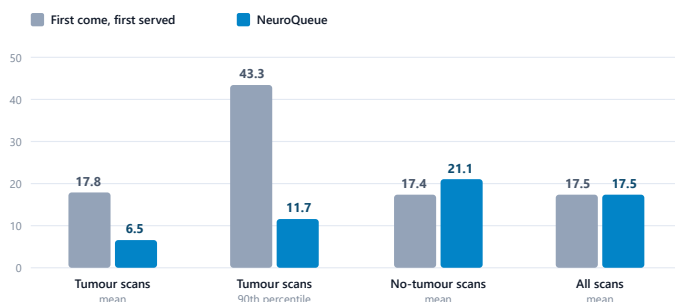
Answers questions about the site by text or voice; never medical advice.

06

Business Value

Results and validation

MINUTES BEFORE A DOCTOR OPENS THE SCAN



6.5 min

Mean wait for a tumour scan

down from 17.8. 90th percentile 11.7, down from 43.3

17.5 min

Mean wait for all scans

unchanged: same doctor, same day, better order

Simulated working day using the model's real held-out predictions: 1 radiologist, 9 scans per hour, 6 minutes per read, 200 simulated days. Not clinical outcomes. Normal scans wait 21.1 minutes instead of 17.4.

TIME

The doctor edits and signs a finished draft instead of typing a report from nothing.

RISK

11 of 12 model errors were still routed for early attention. Contradicting report text is blocked.

PATIENT EXPERIENCE

A plain-language report in their own portal, in English or Arabic.

ACCOUNTABILITY

A hash-chained audit log that shows any later edit or deletion.

Who would adopt it: imaging centres and hospital radiology departments, as a worklist-prioritisation and reporting-assistance layer.

07

Risks & Next Steps

Controls already built, honest limits, and the future scope

RISK	CONTROL ALREADY IN THE PRODUCT
The model is wrong	A doctor reads every scan. Low-confidence and close-call predictions go to Review. An override is one click and is logged
A tumour scan is ranked Routine	Routine needs 99% confidence in no tumour. Anything waiting over 60 minutes is escalated to Urgent. Routine means read last, never cleared
The language model invents a finding	It receives only the doctor's confirmed fields. A rule-based check compares the text against them, and forbidden phrases stop the draft
A patient sees someone else's data	Access is enforced on the server by role and ownership. Patients receive finalized reports only, never model output
A doctor opens scans that are not theirs	Each scan is assigned to one doctor; the server refuses every other doctor
Records are changed afterwards	Append-only audit log where each entry stores the hash of the one before it

HONEST LIMITS TODAY

- Trained and tested on one public dataset of 2D images
- Four classes only; no malignancy or grading claims
- Not a medical device and not clinically validated
- The waiting-time result is a simulation
- Demo hosting, not yet set up for regulated health data

BEFORE REAL CLINICAL USE

- **Validate** on other scanners and sites with radiologist-labelled data
- **Measure** real time-to-read in a reader study
- **Comply:** regulated hosting and the regulatory pathway

Future scope

1 PACS and RIS integration

Pull scans directly from the hospital's imaging system and push the signed report back, so nobody uploads anything by hand.

2 Arabic reports

The interface is already bilingual; next, the PDFs and the patient summary in Arabic too.

3 Notifications

An SMS or email to the patient when their report is ready, and an alert to the doctor when an Urgent scan is assigned.

4 **Beyond brain tumours**

Today it handles brain tumours. The same queue-and-report flow can serve any tumour type with a model trained for it.

5 **Tumour segmentation**

Outline the tumour and measure its size automatically, so the doctor confirms a measurement instead of typing one.

6 **Mobile app**

Patients view their report on their phone; doctors get urgent alerts.

08 **Team & Ownership**

Team Import Innovation: members and responsibilities

VA

Veera Siva Abhishek

Team lead

Backend with Sanmeet · Google sign-in and account access · portal design and workflows after login · English and Arabic experience · chatbot and voice assistant integration

S

Sanmeet

AI model and backend

Dataset, training and evaluation of the model, and the backend with Abhishek

VA

Veera Siva Abhiram

Web application

The web interface and its animations

MA

Mohamed Abubakker

Audit

The audit trail inside the web application

The whole team owns the product decisions and the safety rules described above. Contact: 2200031362csehh@gmail.com · +971 528813637